

Muhammad Awwal bin Mukhzar

IT Infrastructure & Network Engineer

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PROFESSIONAL SUMMARY

IT Infrastructure Engineer with 5+ years of hands-on experience across enterprise network operations, firewall governance, disaster recovery, and security infrastructure — currently supporting critical trading infrastructure at Rakuten Trade Sdn. Bhd. under Securities Commission Malaysia and Bursa Malaysia compliance. Proven track record handling market feed failures, power trip recovery, and live incident triage. Hands-on with FortiGate, Sophos, KEMP WAF, Peplink SD-WAN, Active Directory, and Cloudflare. Aligned to ISO 27001, NIST CSF, ITIL 4.

KEY METRICS

5+	100+	<5 min	<30s	200+	90+
Yrs Experience	Infra Nodes	DR RTO	ISP Failover	AD Accounts	DR Servers

PROFESSIONAL EXPERIENCE

IT Infrastructure Engineer (Morning Shift Lead)

Aug 2023 – Present

Rakuten Trade Sdn. Bhd. | Kuala Lumpur, Malaysia | Regulated Financial Environment (SC Malaysia / Bursa Malaysia)

- Configured IPSec VPN tunnel between Rakuten Trade and Alibaba Cloud for secure cloud migration — defined IKEv2 Phase 1/2 parameters, encryption policies, and traffic selectors; validated end-to-end encrypted connectivity; documented as auditable migration runbook.
- Participated in Bursa Malaysia half-day DR Exercise as part of cross-functional team (IT Infrastructure, IT Governance, Security, Development, Business Users) — contributed IPSec tunnel routing via N2N DR line; validated inter-VLAN connectivity across 90+ DR servers (FO/MX/BO/DB); executed DB snapshots for rollback readiness; supported post-mortem with IT Governance; produced structured remediation checklist aligned to ISO 27001 A.5.29/A.5.30 and ISO 22301 BCM lifecycle.
- Detected and triaged live Log4Shell exploit attempt (CVE-2021-44228) — analysed malicious JNDI LDAP callback intercepted by KEMP WAF/OWASP CRS 3.3.8; escalated to security team; coordinated patch validation across Java/Log4j backend stack; full remediation within same business day with zero business impact.
- Governed FortiGate and Sophos firewall policy lifecycle across Production and DR — led Sophos Firewall Migration and DR Fortinet Firewall Migration; annual security audit eliminating 100% of orphaned rules; zero-downtime policy migrations affecting 50+ rules; policy matrix produced for audit trail (ISO 27001 A.8.20).
- Performed server hardening and enforced least privilege access principles across production environments — conducted periodic access control audits sweeping Active Directory, ticketing systems, and physical server room door access systems.
- Resolved market feed failure before market open — rapidly raised incident, engaged directly with market feed vendor, restored trading data feed within critical pre-market window ensuring no trading disruption.
- Architected Cloudflare DNS failover with orange-cloud proxying strategy, achieving sub-5-minute RTO for critical trading platform — meeting regulatory DR obligations (ISO 27001 A.5.29).
- Deployed DR Monitoring Stack (Icinga2, Grafana, Prometheus) — real-time observability for 20+ infrastructure nodes; automated alerting reduced MTTD by ~40%.

- ▶ Set up Docker containerisation environment and deployed OpenCTI Cyber Threat Intelligence platform via Docker Compose — configured connectors, data ingestion pipelines, and persistent volumes for proactive threat visibility.
- ▶ Utilising Kali Linux on VMware Pro Workstation for penetration testing practice and vulnerability analysis — building offensive security capability to complement defensive infrastructure role.
- ▶ Led Active Directory Tech Refresh — privileged access migration for 200+ accounts with zero authentication downtime using staged DNS cutover approach (ISO 27001 A.8.2/A.8.3).
- ▶ Implemented Peplink SD-WAN Leased Line Failover — automatic ISP failover in <30 seconds; produced DR runbooks and coordinated annual DR tests with 4+ service partners.

IT Services Support Engineer | NOC Engineer

Dec 2019 – Jul 2023

Mesiniaga Berhad — Deployed at Astro Malaysia & SPRM HQ | Kuala Lumpur, Malaysia

- ▶ Monitored ISP Core Network NOC operations covering IP Transit, BGP routing stability, and last-mile connectivity for Astro's national broadband subscriber base — using Huawei NCE, FortiAnalyzer, and ELK for log analysis.
- ▶ Managed ITIL-aligned incident and fault management workflows via Freshdesk and internal ticketing — maintaining SLA compliance for P1/P2 network incidents across a 24/7 NOC environment.
- ▶ Executed Residential Gateway (RGW) provisioning and firmware upgrade campaigns, processing 500+ device activations per quarter with <1% rollback rate.
- ▶ Ensured TM regulatory PMW (Proactive Monitoring Window) compliance — coordinating scheduled maintenance windows and minimising customer impact during planned outages.
- ▶ Produced structured incident reports and escalation documentation, contributing to a 15% reduction in repeated incident categories through root cause trending.
- ▶ Deployed at SPRM HQ — supported IT infrastructure, network operations, and helpdesk escalations in a government security agency environment.

TECHNICAL SKILLS

Network	FortiGate, Sophos Firewall, Peplink SD-WAN, Cisco, Cisco ISE, IPSec/SSL VPN, BGP/OSPF, VLAN, STP, EtherChannel LACP, Subnetting, Cloudflare
Security	KEMP WAF/ModSecurity, OWASP CRS, CVE Analysis, Kali Linux, Server Hardening, Least Privilege Enforcement, Vulnerability Assessment
Containers	Docker, Docker Compose, OpenCTI (Cyber Threat Intel), Container Networking
Systems	RHEL, Windows Server, Active Directory, Nginx, Veeam, Sails.js/PM2/Node.js, VMware Pro Workstation
Monitoring & NOC	Icinga2, Grafana, Prometheus, Huawei NCE, FortiAnalyzer, ELK, Infoblox DNS, Zabbix, SNMP, Syslog, NOC Operations
Cloud & DR	Cloudflare, Azure Fundamentals, IPSec Tunnel (Alibaba Cloud Migration), DR Planning, BCP, ISO 22301 BCM
ITSM	ITIL 4, ISO 27001:2022, Incident/Change/Problem Mgmt, SLA/OLA Compliance, Rotational Shift Operations
Tools	Burp Suite, Ansible, Git, Jira, ServiceNow, Freshdesk

SOFT SKILLS

Soft Skill	Demonstrated Through
Communication	Presented EOL write-offs (RM115K) to management; coordinated across infrastructure, dev, governance, business teams
Leadership	Morning Shift Lead — daily ops, escalation handling, handover coordination, incident command on rotational shift
Problem-Solving	Market feed failure pre-market open, power trip recovery, ORMS triage, server room cooling crisis — resolved under live pressure
Analytical Thinking	Root-cause from WAF/firewall logs, CVE triage, firewall audit, Bursa Malaysia DR post-mortem & remediation planning
Attention to Detail	Zero-error AD migration (200+), DR server validation (90+), periodic AD/ticketing/server door access audits
Teamwork	Cross-functional: Infrastructure, Governance, Security, Dev, Business Users, 4+ external vendors

CERTIFICATIONS

- Fortinet NSE 1, NSE 2, NSE 3 — Network Security Associate | Fortinet | 2024
- Fortinet FCP / NSE 4 — In Progress (2025)

EDUCATION

Bachelor of Computer Science (Networking & Cybersecurity)

Universiti Teknologi Malaysia (UTM) | Expected 2028

Diploma in Digital Technology

Politeknik Ungku Omar | 2017 – 2020

References available upon request